Nat Moges

CREATIVE TECHNOLOGIST · SYSTEMS ARCHITECT

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Master Record v5.3 · updated November 10, 2026 · replaces v5.2 (October 12, 2026)

> **WHAT THIS DOCUMENT IS**
>
> This is the full record, not the one-page résumé. It exists because a one-page document compresses eight years of self-directed work into thirty lines, and compression is what makes the work hard to verify. Everything here is decomposed: the systems, the components underneath them, the numbers with their sources, the parts I adapted from other people, and the limits of every claim. If you received a one-page résumé and wanted to know whether it was real, this is the document that answers that.

SUMMARY

Creative technologist and systems architect. I build the technical systems behind creative output — production infrastructure, AI-native operating systems, and interactive web — and I work the craft tools they serve. Self-hosted infrastructure in continuous service since June 2023, a three-suite relational operating system audited to closure and subjected to external adversarial review, a portfolio with live 3D architected and shipped solo in Framer at natmoges.com, and fourteen React + TypeScript components rebuilt from that portfolio between October 5 and November 4, 2026, each live with a public repo. Every system below was designed, deployed, and maintained by one person with an Associate's degree and zero external capital.

AT A GLANCE

Since June 2023 — self-hosted infrastructure in continuous service, no external support

15+ — Docker services in production on a single consumer host

14 — React + TypeScript components shipped Oct 5 – Nov 4, 2026, each live at a URL with a public repo

3 — core findings demoted by an external adversarial review of the operating system, all accepted

40,516 — YouTube views (YouTube only) · **18,000+** platform plays · **~150** assets produced end to end

3.84 — GPA, graduated With Highest Honor

EXPERIENCE

Creative Technologist & Systems Architect — NatFTP (Independent Practice)

2023 – PRESENT · Philadelphia, PA

- **Designed, built, and audited a relational operating system across three suites** — cross-suite relations, formulas, rollups, and versioned changelog discipline under a written version-issuance rule. Audited to closure and subjected to an external adversarial review that demoted three of its own core findings, all of which were accepted and rebuilt around.
- **Made the system AI-native:** MCP-wired into production, agent-readable and agent-writable, with a specified agent layer and a single-source-of-truth configuration database governing operating parameters.
- **Architected a multi-model AI system on a persistent substrate** — models routed by task and constraint against one source of truth, so a second model executed a full database operation with no handoff document and no context transfer. Constrained generation with provenance rules and blocking verification gates, and a human at every irreversible step.
- **Rebuilt fourteen components from my own portfolio in React + TypeScript on Next.js** — each from a credited reference, each improving on it, each live at a URL from a public repo with one commit per step. Built with an AI co-pilot from plain-language specs I wrote first. Four of them now run on natmoges.com. Detailed below.
- **Architected and shipped natmoges.com solo in Framer** — Spline 3D orbital viewport, working audio terminal, mobile-first architecture. Generated an inbound referral from a Starlink engineer via cold outreach anchored on the portfolio.
- **Architected a multi-tier private cloud on a single consumer host** — a RAID 5 mass-storage pool and a RAID 0 NVMe performance tier serving different access patterns, a Docker control plane behind a WireGuard perimeter with an internal

reverse-proxy routing layer, container lifecycle automation, a virtualized automation domain, and UPS-backed graceful shutdown with external cold backup. Documented as a public architecture brief a stranger can read.

- **Designed the platform layer to be governable rather than hand-tended** — a differentiated update policy set by blast radius: interconnected services version-pinned and manually gated, standalone services on automated updates, a single orchestration surface, and per-service routing. This is what makes multi-year uptime possible for one operator with no on-call rotation.
- **Rebuilt a brand identity end to end using generative AI** — Adobe Firefly, Midjourney and multiple video-generation tools integrated with Adobe Creative Suite craft, shipped and live across web, video and social.
- **Built and ran PRODBYNAT, a solo digital media and product engine** — 40,516 YouTube views, direct-to-consumer e-commerce via BeatStars, driven by SEO and end-to-end media production out of a treated broadcast-grade studio I designed and built: signal flow, acoustic treatment, and multi-camera, multi-destination streaming architecture.
- **Practice answer-engine optimization** — structuring content and site architecture so AI systems parse, retrieve, and cite it correctly.

Client Engagement & Sustainability Intern — Xponance, Inc.

SUMMER 2022 · Institutional buy-side fiduciary asset management · Philadelphia, PA

- Supported client engagement at an institutional buy-side fiduciary asset manager serving public pension systems, corporate treasuries, endowments, and foundations.
- Reported directly to the Managing Director heading Client Engagement & Sustainability; synthesized client-forum and investment-meeting content into structured reports for sales leadership.
- Maintained CRM data integrity and client reporting for institutional accounts using Excel and internal financial and CRM systems.
- Worked inside a multi-manager platform whose function is making underestimated investment managers legible to institutional gatekeepers — learning the consultant layer that gates pension and endowment allocation from the inside.

THE BRIDGE — WHY THIS INTERNSHIP IS LOAD-BEARING

The client-forum notes I took in this role were, structurally, relational CRM architecture drawn freehand at twenty: entities with statuses, contacts with relationship metadata, action items with owner and deadline columns, dated follow-up triggers, strategic reads in italics, knowledge gaps flagged inline. Nobody assigned that structure. It is the direct antecedent of the three-suite relational system I build on now — which makes the operating system a continuation, not a pivot.

Electronics Specialist, Technical Consultant — Target

2020 – 2021 · Philadelphia, PA

Primary technical resource for the Electronics department; resolved 20+ customer technical issues daily and translated product capability into purchase decisions.

Creative Producer, Photo & Video — Independent

2019 – 2021 · Philadelphia, PA

Managed full-lifecycle client photo and video projects from consultation to delivery — on-location direction, professional camera, lighting and audio, drone and real-estate imaging, event coverage, and post-production in Adobe Creative Suite.

EDUCATION

Associate in Arts, Business — Community College of Philadelphia

2022 · Graduated With Highest Honor · 3.84 GPA · 63 credits

Relevant coursework: Risk Management & Insurance · Statistics I · Financial Accounting · Managerial Accounting · Principles of Economics (Macro & Micro) · History of Capitalism.

Risk Management & Insurance was formal coursework with eight graded assignments — risk taxonomy applied to the 2007–09 crisis, measures of dispersion and the diversification tradeoff, pre-loss and post-loss objectives, actuarial fundamentals including loss and combined ratios and IBNR reserving, and pension mechanics including the DB-to-DC shift. Stated at its honest altitude: introductory undergraduate coursework, not professional risk analysis.

THE CLOSER LADDER — REACT + TYPESCRIPT, OCTOBER 5 – NOVEMBER 4, 2026

Until October, my web work lived in Framer, and every listing that asked for React and TypeScript found nothing behind the claim. So I took my own site apart and rebuilt every sourced component in code, in order: one component, then data, then the page, then the pixels, then 3D. Each build reuses the last and adds exactly one new idea. Fourteen builds later, the polished front end has a server side of its own: the record at record.natmoges.com reads its data from Notion on the server.

Method. Before any code, I write a plain-language paragraph covering every moment of the component's life — before the user acts, on the action, during, at the end, and when the browser refuses. That paragraph stays above its code as a comment. An AI co-pilot (Claude) writes code from the paragraph and explains it in my terms; I name every variable and container, make the design calls, and check every change before it's committed. One commit per step, so the git history is the learning record. Every reference author is credited in the README, and none of their code is in the repo.

Since November 10, 2026, four of the builds run back on natmoges.com as Framer code components: the case-study folders (04), the motion tiles (05), the rocket (03) and the smooth scroll (08).

01 · vinyl — a record player

Oct 5, 2026 · vinyl-swart.vercel.app · github.com/natmoges/vinyl · reference: DiscPlayer by Ahmad

A real button with three states — playing, paused, and blocked when the browser refuses playback. The disc angle is computed from the song's position, so scrubbing turns the record both ways. New idea: state, refs and an animation-frame loop.

02 · TextScrambler — a signal fighting through noise

Oct 6, 2026 · record.natmoges.com/textscrambler · reference: TextScrambler by Solt Wagner

Characters resolve from noise left to right, with a speed control that works and the real word exposed once to screen readers. New idea: props and a TypeScript interface — the first reusable component.

03 · Back2Top — a rocket that rides you back to the top

Oct 12, 2026 · record.natmoges.com/back2top · reference: BackToTop by The Velox Studio

The ride's length follows the page as a speed with a cap, any wheel, touch or key breaks it, and a hidden button is truly gone. New idea: listening to the window.

04 · AnimatedFolder — folders that fan open from data

Oct 14, 2026 · record.natmoges.com/animatedfolder · reference: AnimatedFolder by Launchly

One component, three copies, three lists. It fixed a trapped-viewer bug my site had, and since Nov 10 the build itself runs there. New idea: lists from data and children.

05 · Motion Tiles — seven video tiles, one playing at a time

Oct 26, 2026 · record.natmoges.com/motiontiles · reference: Motion Tiles by Uzair J.

My 2019–2022 arc as a tilted 3D stack of my real photographs. New idea: many refs at once.

06 · SignalLocation — where you're reading from

Oct 31, 2026 · record.natmoges.com/signallocation · reference: Live Location by Amr Rashed

The reader's location from my own route, fresh every visit, five states on one tempo. New idea: async data with loading, error and fallback states.

07 · Accretion — my record, read live from Notion

Nov 1, 2026 · record.natmoges.com · no reference

The record scanned in order: kept if the record can prove it, debris if it can't yet. The first build with nothing to copy. New idea: server-side data.

08–14 · the page, the pixels, 3D

Nov 1 – Nov 4, 2026 · all on record.natmoges.com

08 SmoothScroll (wrapping Lenis; ref: Framer University) · 09 TrailCursor (ref: Akın Gündoğan) · 10 LogoPreloader (ref: Brett Jackson, Sukoya Design) · 11 StarCloud, 440 particles on one canvas (ref: Sabo Sugi) · 12 Gravity, a satellite in a physics engine (ref: Framer University) · 13 BootScreen, the HUD powered on like a CRT (ref: Blue Jackson) · 14 HeroOrbit, the planet in WebGL (ref: Tim Quirino's Orbital View of Arrakis).

Every build was checked live from outside my machine and on iPhone Safari. The honest boundary on this work is stated under What I do not claim.

THE THREE SYSTEMS

Each of these is published at natmoges.com as a case study structured Artifact, Breakdown, and Toolchain. Summarized and corrected here so this document stands alone; where the record has since sharpened a published sentence, this version is the current one.

I. Sovereign Infrastructure — a private cloud on consumer hardware

Constraint. Commercial cloud solutions failed on three non-negotiables: cost at scale, latency for real-time creative workloads, and full data sovereignty. With zero capital and a bedroom studio, I had to design a private cloud that delivered reliable, multi-year operation on consumer hardware.

Architecture. A multi-terabyte server environment on a QNAP TS-464. Storage is deliberately two-tiered by access pattern: a ~60TB RAID 5 array for redundant mass storage and sequential-read media, and a ~7.2TB RAID 0 NVMe array for low-latency

active project work. Application deployment runs on Docker orchestrated with Portainer, isolating an automated media pipeline, with a torrent client confined to a VPN-isolated network namespace. Nothing is exposed to the internet: remote access is a WireGuard tunnel into the network, so the VPN is the perimeter, and an NGINX reverse proxy sits behind it as the internal routing and naming layer — services get hostnames instead of ports. A dedicated virtual machine runs Home Assistant OS. That move was a physical-to-virtual migration of a stateful service, and the requirement was set before the method: nothing gets configured again on the far side. The device pairings and history were the asset and could not be re-derived, which forced an image-level migration over a rebuild. It was operational on first boot, and it sandboxes event processing away from the service plane so an automation failure cannot take down media or storage.

Result. In continuous service since June 2023, with the container platform since February 2024, and no external support. The array serves as the single source of truth for creative and business files. The uptime is an outcome of the resilience design, not of endurance.

II. Digital Product Engine — PRODBYNAT, 2019–2021

Constraint. Entering the hyper-competitive music production marketplace with zero capital, zero team, and zero distribution network required building a self-contained product engine capable of competing with funded studios.

Architecture. A solo pipeline I ran end to end: every asset moved from a raw MPC/DAW session, to an After Effects 3D visualizer, to SEO metadata, to the storefront. It had handoffs — I was all of them. The system was built around a content-first distribution funnel using YouTube as the primary driver for audience acquisition. Templates and batch processing carried ~150 digital assets through it, including custom 3D audio-visualizer rendering and metadata across every asset. The files carry a strict naming schema, Name_Key_BPM: the first two beats predate it, and 123 of the 126 packaged beats match it exactly.

Result. Direct-to-consumer sales through a BeatStars storefront, 40,516 YouTube views, 18,000+ platform plays and 160+ subscribers — while stress-testing the operational principles later codified in the business operating system.

The output of a pipeline is not throughput. It is the removal of a decision. Every template and batch process eliminates a choice that would otherwise be made again for each asset, which is why ~150 assets shipped solo was possible at all.

III. Business Operating System — three suites, relational, MCP-native

Constraint. After scaling PRODBYNAT from bedroom project to revenue engine, the cognitive overhead of managing strategy, execution, and creative work in parallel became the next bottleneck. I needed an external system that mirrored non-linear thinking while enforcing operational rigor.

Architecture. A network of relational databases across three suites — life, work and business — modeling every core function: projects, tasks, content calendar, CRM, financials, and a library of core templates and standard operating procedures. Strategic frameworks are built into the system rather than kept beside it: a Business Model Canvas, a Risk Register, and a "Curiosity Threads" database for synthesizing non-linear insight into actionable strategy. The system is organized in three layers by altitude of attention: a Command Center for strategic oversight, a Knowledge Base of reusable process, and a Project Hub for granular architecture and execution. A single configuration database holds the operating parameters, and a claims file governs what may be said about the work: every number traces to a source, and every authorship adjective must name its sourced parts.

Result. A single source of truth that translates strategic intent into executable workflows, readable and writable by AI agents with a human at the irreversible step. The load-bearing interface decision is a context-specific view called the Creator Cockpit, which surfaces only the tools and tasks relevant to creative work — deliberately separating execution mode from planning mode.

INFRASTRUCTURE, DECOMPOSED

"Self-hosted infrastructure" is one phrase covering many decisions. This is what is actually underneath it.

HOST

QNAP TS-464 · Intel Celeron N5105 · 16GB DDR4 · ~$600 consumer hardware. The constraint is the point. I picked that chassis because its two M.2 slots don't cost a drive bay — the upgrade path was known before the box was bought.

NETWORK AND EDGE

eero 6+ gateway · TP-Link 8-port 2.5GbE switch · flat trusted LAN zone · 1500VA UPS. Remote access only through a WireGuard tunnel; no service is published to the internet.

STORAGE AND DATA GOVERNANCE

Two pools serving different access patterns: RAID 5, 4 × 20TB, ~60TB usable for redundancy and mass storage; RAID 0, 2 × 4TB NVMe, ~7.2TB usable for cache and active projects. Shares split by purpose, with access patterns designed rather than inherited — media library sequential-read, backup target write-heavy, active projects low-latency. Container state lives on bind mounts to the array, where the snapshots and the cold copy already reach, not inside Docker's storage graph where the protection layer can't see it.

RESILIENCE AND RECOVERY

Coverage is mapped to failure modes: RAID covers a dead disk, snapshots cover a bad delete, the UPS covers a mid-write power cut, and the external cold copy covers the building.

WORKLOAD PLACEMENT

The server is a Celeron because decode runs on the client. Direct play means the NAS reads sequentially and pushes bytes. Moving playback to a client app removed the media server from the architecture entirely — a performance problem solved by deleting a component rather than upgrading one.

VIRTUALIZATION

An autonomous domain running as a guest VM: a residential control plane across five protocol families — Zigbee, MQTT, cloud APIs, local network control and HomeKit — through one orchestrator, over a spatial entity model I built. The automations carry tuned hysteresis, conditional guards, and proxy sensing where a room had no reliable vacancy signal.

CONTAINER PLATFORM

Routing via NGINX Proxy Manager behind WireGuard. Governance and lifecycle via Portainer orchestration, a single dashboard, and an updater that runs on demand under the blast-radius policy: every image tracks latest except one, pinned to an exact build, and the pin and the updater's exclusion list are the same policy expressed twice. Application workloads include an automated media suite and a client inside a VPN-isolated network namespace. Hardening is mostly absence: eight of thirty available services are on, because surface that isn't present doesn't need defending.

TECHNICAL CAPABILITY

WEB & INTERACTIVE

Framer (production environment) · React + TypeScript on Next.js (App Router, client and server components, CSS Modules; fourteen components shipped — see boundaries) · Framer code components · git, one commit per step · Vercel deployment from main · Spline 3D · WebGL · responsive and mobile-first architecture · interaction design · accessibility basics (real buttons, aria labels, focus handling, reduced motion) · HTML/CSS

SYSTEMS & INFRASTRUCTURE

Self-hosted NAS (QNAP TS-464) · Docker and Container Station, 15+ services in production · virtualization and VM management, including P2V migration of a stateful service · WireGuard VPN perimeter and VPN-isolated network namespaces · NGINX reverse proxy as internal routing · multi-tier storage architecture (RAID 5 and RAID 0 NVMe) · UPS-backed graceful shutdown, snapshots and external cold backup · update policy by blast radius · 2.5GbE networking · SMB/AFP shares and access-pattern design · CLI-level Linux administration and troubleshooting

AI & AUTOMATION

MCP integration in production · agent-readable and agent-writable system design · multi-model orchestration on a persistent substrate · constrained generation with provenance rules and blocking verification gates · context and instruction architecture · AI-co-piloted software development from written specs · answer-engine optimization · AI-assisted production workflows

DATA & SYSTEMS ARCHITECTURE

Relational database design · cross-suite relations, formulas and rollups · versioned changelog discipline · audit methodology · provenance classification of claims and figures · CRM data integrity · information architecture · knowledge management and SOP design · Notion (system architecture) · Excel

INTERFACE & PROCESS DESIGN

Context-specific interface design for operator tools · information hierarchy across attention altitudes · systems design and process engineering · repeatable workflow design · batch processing and pipeline architecture · phased sprint execution and granular task tracking

MEDIA PRODUCTION

Adobe Creative Suite (Premiere Pro, After Effects, Photoshop) · generative AI for image and video (Adobe Firefly, Midjourney, video generation) · Final Cut Pro · multi-camera systems · cinematic lighting · OBS and multi-destination streaming · drone and real-estate imaging

AUDIO

Logic Pro · MPC Live III, standalone end-to-end · FL Studio · Universal Audio Apollo Twin X · Shure SM7B · professional signal flow · treated-room studio design

BUSINESS & OPERATIONS

E-commerce and digital product distribution · SEO and content strategy · agile methodologies · client engagement and reporting in an institutional context

WHAT I DO NOT CLAIM

Most résumés list only capability. This section exists because a document that states its own limits is easier to verify than one that does not — and because knowing where a claim stops is what makes the rest of it usable.

React and TypeScript proficiency. What I claim is narrow and checkable: I built and deployed fourteen React + TypeScript components on Next.js, with an AI co-pilot, and the READMEs say so. I do not claim proficiency or years with the stack, production front-end systems, component libraries or design systems in code, testing, state libraries, team workflows (branches, pull requests, code review), or writing the code without assistance. Framer remains the production environment for natmoges.com.

Custom code on natmoges.com. Four of my own builds run there since November 10, 2026, ported into Framer code components: the case-study folders, the motion tiles, the rocket and the smooth scroll, plus its 404 page. The rest is composed. The hero is an adaptation of Tim Quirino's masthead at timq.xyz, and the other components come from community authors, each credited. What is mine beyond those builds is the information architecture, the token system, the breakpoint strategy, the copy, the selection and the wiring.

Machine learning engineering or research. I am an AI systems integrator and orchestrator. I do not train models, fine-tune them, or work in ML theory, evaluation methodology, or inference optimization.

SRE, DevOps, or platform engineering. What I run is a single-host private cloud on consumer hardware. No Kubernetes, no multi-node high availability, no formal monitoring or alerting stack, no CI/CD for the infrastructure. Infrastructure-as-code is partial: four of fifteen services have a declarative definition, and the other eleven would need rebuilding by hand. I do not claim certificate lifecycle management.

Multi-agent deployment in production. The agent layer in my operating system is specified, not shipped. What is shipped is MCP-native orchestration with a human at every irreversible step.

Generative AI at daily production volume. What I have is shipped brand and identity work built with generative tooling inside a real system — not a daily generative production pipeline.

Professional risk analysis. Formal undergraduate coursework in risk management and insurance. Materially more than following markets; materially less than pricing risk professionally.

Deep instrumented analytics. GA4, Search Console, and formal Core Web Vitals work is my lightest area.

Years in title. I do not claim four to seven years as a titled creative technologist. The work is longer than the title; the title is recent because the category is recent.

Enterprise team leadership. Every system here was designed for one operator and has never been run by a team. It was built so it could be handed to one — that is design intent, not a demonstrated result.

HOW TO VERIFY ANY OF THIS

- **natmoges.com** — the portfolio, architected and shipped solo in Framer, with all three case studies published at Artifact, Breakdown, and Toolchain depth, and the readable résumé at natmoges.com/resume.
- **record.natmoges.com** — the public record, read live from Notion: what the record can prove is kept, what it can't yet is debris.
- **github.com/natmoges** — the public repos for the React + TypeScript builds (vinyl, component-rebuilds), with commit histories one step per commit, READMEs crediting every reference author and the co-pilot, and live deployments linked above.
- **natmoges.com/recall** — Recall Is Not Evidence, an essay in which I audited my own claims against primary sources: filesystem timestamps, platform ledgers, a 2020 notebook. More than twenty corrections, in both directions.
- **The architecture brief** — a complete system diagram of the infrastructure, including host specifications, storage layout, network topology, and every service in production.
- **References** — available on request, including the Managing Director I reported to directly at Xponance.

Every number in this document is drawn from a maintained metrics ledger with a recorded source and a provenance class. Two figures that appear on my portfolio are deliberately excluded here because they are self-measured without recorded methodology; I will state them in conversation with that qualifier attached, but I will not print them as evidence.

THE ONE-PAGE VERSION

A one-page résumé tailored to a specific role is available and is what I submit with applications. This document is the record behind it. If you have both, the one-pager tells you what I would do in your role, and this tells you whether the one-pager is true.